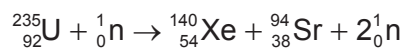


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6. (a) One possible fission reaction of uranium-235 (U-235) is



The following data are available.

Mass of one atom of U-235 = 235 u

Binding energy per nucleon for U-235 = 7.59 MeV

Binding energy per nucleon for Xe-140 = 8.29 MeV

Binding energy per nucleon for Sr-94 = 8.59 MeV

- (i) State what is meant by binding energy of a nucleus.

[1]

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- (ii) Outline why quantities such as atomic mass and nuclear binding energy are often expressed in non-SI units.

[1]

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- (iii) Show that the energy released in the reaction is about 180 MeV.

[1]

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- (b) A nuclear power station uses U-235 as fuel. Assume that every fission reaction of U-235 gives rise to 180 MeV of energy.

- (i) Estimate, in J kg^{-1} , the specific energy of U-235.

[2]

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(Question 6 continued)

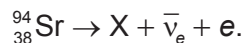
- (ii) The power station has a useful power output of 1.2 GW and an efficiency of 36 %. Determine the mass of U-235 that undergoes fission in one day. [2]

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- (iii) The specific energy of fossil fuel is typically 30 MJ kg^{-1} . Suggest, with reference to your answer to (b)(i), **one** advantage of U-235 compared with fossil fuels in a power station. [1]

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- (c) A sample of waste produced by the reactor contains 1.0 kg of strontium-94 (Sr-94). Sr-94 is radioactive and undergoes beta-minus (β^-) decay into a daughter nuclide X. The reaction for this decay is



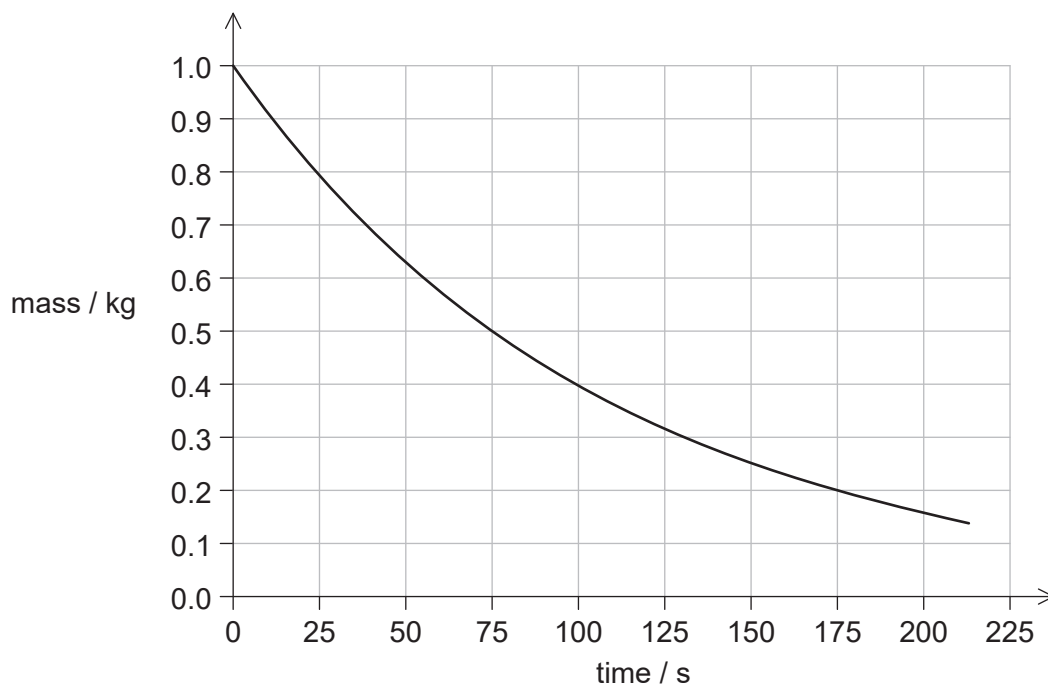
- (i) Write down the proton number of nuclide X. [1]

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(Question 6 continued)

The graph shows the variation with time of the mass of Sr-94 remaining in the sample.



(ii) State the half-life of Sr-94.

[1]

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(iii) Calculate the mass of Sr-94 remaining in the sample after 10 minutes.

[2]

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